11 Explicit Extensions: Not the Baseline

Separation rule. Every object in this section is inactive in the baseline. None enters its symbols, assumptions, equilibrium, propositions, or empirical mapping. Activating any block would require a distinct model specification with its own assumptions, proofs, and data justification.

11.1 Extension 1: Smooth route choice

Let W_{ir} denote the deterministic value of route r, in currency per project. In this inactive extension only, let ϵ_{ir} be an additive route shock and U_{ir} its perturbed value. If a future quantitative application requires smooth route shares, it may use

$$U_{ir} = W_{ir} + \epsilon_{ir}, \qquad r \in \{I, E, T, A\}.$$

Under an independently justified iid Type-I extreme-value distribution with scale $\sigma > 0$, the extension-only probability would be

$$P_{ir}^{\text{logit}} = \frac{\exp(W_{ir}/\sigma)}{\sum_{\ell \in \{I,E,T,A\}} \exp(W_{i\ell}/\sigma)}. \tag{1}$$

For completeness, let the shock CDF and density be $G_\sigma(e) = \exp[-\exp(-e/\sigma)]$ and $f_\sigma(e) = G'_\sigma(e)$. Conditional on the shock for route r, independence gives

$$P_{ir} = \int_{-\infty}^{\infty} f_\sigma(e) \prod_{\ell \neq r} G_\sigma(e + W_{ir} - W_{i\ell})\, de.$$

With $t = \exp(-e/\sigma)$, this equals

$$\int_0^\infty \exp\left[-t \sum_\ell \exp((W_{i\ell} - W_{ir})/\sigma)\right] dt = \frac{1}{\sum_\ell \exp((W_{i\ell} - W_{ir})/\sigma)},$$

which is the stated formula. A route with value $-\infty$ receives zero probability by its limiting interpretation; the abandonment option keeps the denominator positive. The exponent is dimensionless because σ and route value use the same units.

This formula is not used by any baseline demand, cutoff, proposition, or equilibrium equation. Activation requires route-share data, a quantitative need, and shock-scale identification. Deterministic r_i^* remains the theoretical baseline.

11.2 Extension 2: Route-specific implementation

If later evidence supports route-specific post-assignment implementation probabilities, an extension may introduce $\chi^I(q, m)$ and $\chi^E(q, m)$. These would replace neither the baseline route choice nor the policy-invariant downstream probability without a new institutional model. In particular,

$$M = 1 \;\not\Rightarrow\; \chi^E(q, m) \uparrow$$

by assumption. Activation requires an operational definition, route-aligned data, and a causal justification for any policy dependence.

11.3 Extension 3: Transfer-market microfoundation

A future adoption-cost, search, or bargaining block may microfound the outside value $T(q, m)$. No bargaining protocol, transfer price, adoption cost, or counterparty choice is selected here. Until a separately specified design specifies agents, information, surplus, outside options and timing, $T(q, m)$ remains the exogenous non-retained outside option of the baseline.

11.4 Extension 4: Dynamic firm evolution

A recursive model is admissible only when a future question and evidence support a genuine state transition, such as accumulated manufacturing capability, a persistent product portfolio, entry/exit, or transition from a research-oriented developer to an integrated incumbent. No Bellman equation is written here because no state vector or transition law has been specified. This prevents static continuation-value accounting from being relabeled as a dynamic foundation.

11.5 Extension 5: Multi-CMO matching

A richer search or matching model may be considered if match-level data identify developers, suppliers, technology compatibility, search duration, prices and match timing. No matching function, bargaining rule or congestion externality is imposed here. The baseline continues to clear one aggregate qualified-capacity market at p_m^*.

11.6 Extension 6: Research-versus-development allocation

An optional, separately specified block may use two controls: upstream research x_i^R and project development x_i^D. A resource or financing constraint might link them. The block could be used to study $x_i^D \uparrow$ together with $x_i^R \downarrow$. The block is not implemented here: no constraint, objective, FOC, patent-production function, or policy sign is specified.

Activation requires a separate decision because it adds a control dimension, can shift the paper toward financial constraints, and overlaps with the empirical mechanism studied by Shi Gu. The baseline retains one common x_i, commercialization organization, retained authorization, manufacturing-capability matching, and CMO scarcity.

11.7 Isolation conclusion

The six optional blocks are documented so that future work has explicit insertion points. They remain inactive and cannot be cited as baseline mechanisms or assumptions. Conversely, the baseline does not import this file or depend on any extension-only symbol.